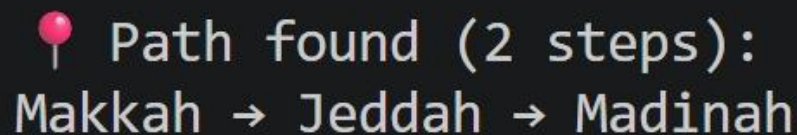
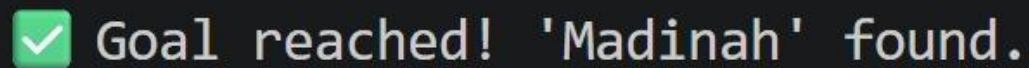


Part 1 – Search (BFS)**Exercise 1 – Change Goal City****Code Modification**

Changed:

```
102     start_city = "Makkah"
103     goal_city  = "Madinah"
```

Output

Answers**a. Number of steps:**

2 steps

b. Path found:

Makkah → Jeddah → Madinah

c. Is it the shortest path?

Yes, BFS always finds the shortest path in terms of number of steps.

Exercise 2 – Add New City (Abha)**Code Modification**

Added:

```
18     "Abha": ["Taif"],
```

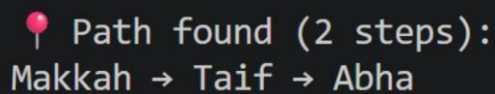
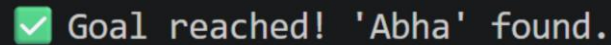
Modified:

```
14     "Taif": ["Makkah", "Riyadh", "Abha"],
```

Changed goal:

```
102     start_city = "Makkah"  
103     goal_city  = "Abha"
```

Output



Answer

Path found:

Makkah → Taif → Abha

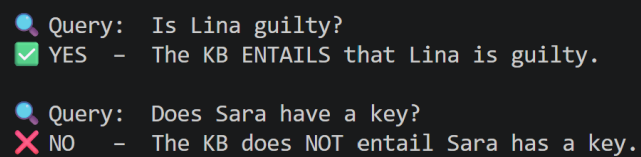
Explanation:

BFS explored neighbors level by level and found the shortest path to Abha.

Part 2 – Knowledge (Logic)

Initial Run

Output



Explanation

The knowledge base proves Lina is guilty because:

- Lina was seen near the room
- Therefore she has a key
- Therefore she is guilty

Exercise 3 – Add Nora

Code Modification

Added:

```
36 nora_key = Symbol("NoraHasKey")
```

```
61 kb.tell(nora_key)
```

Added query:

```
82 nora_guilty = Symbol("NoraIsGuilty")
83 all_symbols.append("NoraIsGuilty")
84
85 answer3 = check_all(kb, nora_guilty, all_symbols)
86 print(f"Is Nora guilty? {answer3}")
```

Output

```
Query: Is Lina guilty?
✓ YES - The KB ENTAILS that Lina is guilty.

Query: Does Sara have a key?
✗ NO - The KB does NOT entail Sara has a key.

Is Nora guilty? False
```

Answers

a. Is Nora guilty?

No

b. Why?

Because there is no rule connecting Nora having a key to being guilty.

c. Missing clue:

```
kb.tell(Implication(nora_key, nora_guilty))
```

Exercise 4 – Remove a Clue

✓ Code Modification

Commented:

```
# Clue 3: Lina WAS seen  
#kb.tell(lina_seen)
```

Output

```
🔍 Query: Is Lina guilty?  
❌ NO - The KB does NOT entail that Lina is guilty.  
  
Is Nora guilty? False  
🔍 Query: Does Sara have a key?  
❌ NO - The KB does NOT entail Sara has a key.
```

Answers

a. Is Lina still guilty?

No

b. Why?

Because removing the clue breaks the reasoning chain.

Without knowing Lina was seen near the room, the system cannot conclude she has a key or is guilty.

Discussion Questions

1. Why does BFS visit wrong cities?

BFS explores all neighboring nodes level by level, so it may visit cities that are not part of the final path. This is normal and ensures the shortest path is found.

2. BFS vs DFS

BFS is better for finding the shortest path because it explores level by level, while DFS may go deep into the wrong path.

3. Better algorithm with distances

Uniform Cost Search (or A*) can be used to find the cheapest path when distances are involved.

4. Implication vs Entailment

Implication is a logical rule (if P then Q), while entailment means that a conclusion is always true based on the knowledge base.

5. Model checking scalability

With 10 symbols, there are $2^{10} = 1024$ possible worlds. This becomes inefficient for large problems.